

VIJAY RATHOD

Full Stack Developer (MERN) | Backend & Frontend Engineer

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PROFESSIONAL SUMMARY

Full Stack Developer with over 3 years of experience designing and building scalable web applications using React.js, Next.js, Node.js, and Nest.js. Skilled in relational database design, REST API development, real-time systems (Socket.IO, RabbitMQ), authentication/authorization (JWT, OIDC, RBAC), and AI/LLM integration. Proven track record of owning features end-to-end — from requirement gathering and system design through development, testing, and production deployment — while mentoring junior developers and coordinating cross-functional teams.

TECHNICAL SKILLS

Frontend: React.js, Next.js, React Hook Form, React Flow, TanStack Query, Electron

Backend: Node.js, Nest.js, Express.js, TypeScript, RabbitMQ, Socket.IO

Database: PostgreSQL, Redis, Sequelize ORM, Schema Design & Migrations

Authentication & Security: JWT, OIDC, Redis Sessions, Refresh Tokens, RBAC

AI / Integration: AI SDK Integration, LLM Integration, LiveKit (Audio/Video Calling)

Other Tools: WebPush, Firebase Cloud Messaging (FCM), REST API Design, Cron Jobs, Git

Other Skills: Agentic AI Development, Prompt Engineering

PROFESSIONAL EXPERIENCE

Full Stack Developer (MERN) — eSparkbiz Technologies

Jan 2023 – Present

- Gather and analyze requirements for new projects and features directly with clients, translating business needs into technical specifications.
- Plan development cycles end-to-end: design database schemas, implement backend and frontend logic, perform testing/validation, and deploy to production.
- Debug and resolve production issues in real time, redeploying fixes with minimal downtime to maintain a seamless user experience.
- Mentor junior developers, support team members, and coordinate cross-functional collaboration to ensure smooth project delivery.

KEY PROJECTS

Smackdab CRM — Automation Module (Full Stack)

- Built the Automation module from scratch, enabling users to design automation flows for CRM activities such as sending emails/SMS and creating or updating CRM entities.
- Designed a relational database schema with foreign-key relationships and implemented schema migrations using Sequelize ORM.
- Developed REST APIs with Node.js, Express, and TypeScript to manage automation creation, updates, steps, and deletions.
- Implemented background processing for automation execution using RabbitMQ queues, plus a cron system for scheduled and delayed automations.
- Built a Socket.IO service to push real-time automation run events to the frontend, keeping the UI updated without manual refresh.
- Created a dynamic node-and-edge automation workflow UI using React Flow, with React Hook Form–driven step configuration synced to the backend.
- Integrated TanStack Query for frontend data caching and real-time updates, with cache invalidation triggered by backend socket events.

Smackdab Auth — Central Identity System (OIDC)

- Designed the initial database structure and schema for a centralized identity system covering organizations, workspaces, and users.

- Implemented login and registration flows using JWT, Redis-backed sessions, and refresh tokens.
- Built the user invitation flow and a Role-Based Access Control (RBAC) system supporting Owner, Admin, and Member roles.
- Implemented protected routes with role- and permission-based access control across the application.

Smackchat — Team Chat & Messaging Platform (Project Lead)

- Currently leading development of Smackchat, a team messaging platform built with React and Nest.js.
- Implemented messaging, replies, reactions, and threaded conversations end-to-end across backend and frontend.
- Integrated Socket.IO for real-time message delivery and updates.
- Implemented WebPush for browser notifications and Firebase Cloud Messaging (FCM) for desktop (Electron) notifications.
- Built and shipped the Electron desktop application for Smackchat.
- Implemented TanStack Query with IndexedDB for efficient client-side data caching.
- Integrated LiveKit for audio and video calling functionality.
- Implemented an auto-translation feature using LLM model integration.

EDUCATION

Master of Computer Applications (MCA)

April 2023

GPA: 8.0 / 10